Preprint

LETTER

Nuclear shell structure and binding energies from tetrahedral packing at $K = 4$

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Abstract

A single algebraic formula, derived from the tetrad geometry of general relativity, reproduces nuclear binding energies across all 83 stable elements from hydrogen to bismuth with zero adjusted parameters. The binding energy per nucleon is $\mathrm{BE}/A = C\,(f - F_{\mathrm{ZERO}})$, where $f(K, s) = 1 + Ks + K^2$ is the coupling polynomial of the stellated octahedron evaluated at nuclear compression, $F_{\mathrm{ZERO}} = K^2 + 1 = 17$ is the Field Origin capacity, and the energy constant $C = m_e c^2\, F_{\mathrm{ZERO}}/K = 2.172$ MeV is fixed by the electron mass and the $K = 4$ topology. Across 82 isotopes from helium to bismuth the formula achieves $R^2 = 0.99$ on total binding energy, with mean error 3.9% and 98% of elements within 10%. A packing algebra based on tetrahedral close-packing with Sierpiński gate exclusion produces all seven nuclear magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ exactly, with $K = 4$ as the unique solution among all integers $2 \le K \le 10$. The nesting ceiling $K^K = 256$ predicts that no shell beyond the eighth can close, consistent with bismuth-209 as the heaviest stable isotope. The same polynomial, evaluated at galactic scales with a single-face coupling path, produces galaxy rotation curves without dark matter.

1 Framework and coupling polynomial

The tetrad is the tetrahedron

The tetrad formulation of general relativity, inherited from Einstein (1928), Cartan (1922), and Weyl (1929), describes the local gravitational frame as a set of four independent basis vectors at each point in spacetime. Algebraically, it is a transformation matrix. Geometrically, four non-coplanar vectors from a common origin define a tetrahedron: the simplest solid enclosing volume in $D = 3$ spatial dimensions, with $K = D + 1 = 4$ faces. The tetrad is the algebra; the tetrahedron is the geometry. They are two descriptions of the same local frame. Regge [3] made the identification explicit: in his discretisation of general relativity, a 3-simplex (a tetrahedron) is the fundamental building block of space, and all curvature concentrates at the vertices where tetrahedra meet.

This identification is not limited to the single-metric case. Hassan and Rosen [4] proved that ghost-free bimetric gravity requires two independent tetrads, e^a_μ and f^a_μ, coupled through an interaction potential. Since each tetrad is geometrically a tetrahedron, the theory requires two tetrahedra at every point in spacetime. Three further results reach the same conclusion: vacuum stability requires opposite orientations to cancel the Boulware–Deser ghost [5]; the $SL(2, \mathbb{C})$ double cover of the Lorentz group assigns one chirality per chamber [6]; and Ashtekar's self-dual/anti-self-dual decomposition [7] assigns one chirality per chamber. Four independent arguments. Two tetrahedra.

Gravity Field Dynamics (GFD) [1] begins here. It takes the algebraic requirement for two interacting tetrads and draws the geometry: what does the interior field structure of a gravitational field look like when both tetrahedra are present?

Two chambers, not one

Consider how gravitational fields couple to their neighbours. In any close-packed field lattice, each field region borders twelve others, the face-centred cubic (FCC) coordination number, established by Kepler in 1611 and proved optimal by Hales in 1998. These twelve neighbours press inward on the central field through shared boundary planes, the same way twelve soap bubbles pressed around a central one flatten its walls into planar facets. The twelve boundary planes are not randomly oriented. In FCC packing, they sort into exactly four groups of three, each group sharing a common normal direction. Four groups, four normals, four faces: a tetrahedron emerges from the packing geometry itself.

But packing is symmetric. The twelve neighbours do not push from one side only; they push from all sides. The four face-normals of the inward-pointing tetrahedron have four partners pointing outward in the opposite directions. These outward normals define a second tetrahedron, inverted with respect to the first. The two tetrahedra interlock with opposite orientation, each vertex of one sitting at the centre of a face of the other. Their compound is the *stellated octahedron*: 8 vertices, 12 edges, 8 triangular faces.

The interior of this compound has three distinct regions. The primary tetrahedral chamber (TetA) is the outward-facing field

structure, the sector that couples to neighbours and carries the massless gravitational mode. The inverted chamber (TetB) is the bounded interior, the sector that can hold confined field content and carries the massive mode. At the geometric centre, all four body diagonals cross at a single point. This is the *Field Origin*: the coupling boundary between the two chambers, the unique location where both tetrads are evaluated at the same spacetime event. It is the interaction vertex of bimetric gravity [4], given geometric location.

On each face of TetA, the opposing vertex of TetB penetrates the surface and subdivides it into $3 + 1 = 4$ sub-regions: three spatial corners and one central core. Across all four faces, this produces $4 \times 4 = K^2 = 16$ independent coupling channels. These channels are the K^2 components of the frame transition matrix in bimetric gravity, now visible as geometric features of the stellated octahedron.

The coupling polynomial

The tetrad equation contains three structural types. A nondegeneracy condition ($\det(e) \neq 0$): one binary constraint ensuring the field closes. A frame (e^a_μ): $K = 4$ independent directions spanning the tangent space. A connection (η_{ab}): a $K \times K$ matrix with $K^2 = 16$ slots encoding how every direction couples to every other. One constraint, four legs, sixteen components. The coupling polynomial counts these structural levels:

$$f(K, s) = 1 + Ks + K^2 \,, \tag{1}$$

where $s \in [-1, +1]$ parameterizes the position between the two chambers. In the three structural regions of the stellated octahedron, the polynomial evaluates to three integers:

$$\begin{aligned} F_{\text{PLUS}} &= f(K, +1) = 21 \;\; \text{(TetA)} \,, \\ F_{\text{ZERO}} &= f(K, 0) = 17 \;\; \text{(Field Origin)} \,, \\ F_{\text{MINUS}} &= f(K, -1) = 13 \;\; \text{(TetB)} \,. \end{aligned} \tag{2}$$

These are arithmetic-locked: $(21 + 13)/2 = 17$, equally spaced by $K = 4$, with no freedom to adjust. The integer $K = 4$ is itself selected uniquely by $2(D{+}1) = 2^D$, whose only positive integer solution is $D = 3$ [1].

The constitutive law $\mu(x) = x/(x{+}1)$ is not postulated as in MOND phenomenology but derived as the Möbius map of this polynomial [1], and its antiderivative yields the Bekenstein–Milgrom energy functional [8] that produces galaxy rotation curves. The Hassan–Rosen interaction potential, whose five free parameters $\{\beta_0, \ldots, \beta_4\}$ are conventionally fitted to data, is locked by the dual tetrahedral geometry to $\{1, 0, K, 0, 1\}$ [1], reducing five constants to zero. The framework is covariant-complete: the polynomial, the constitutive law, the energy functional, and the bimetric parameters all follow from $K = 4$ and the stellated octahedron.

Nuclear scales

In this letter, we apply the coupling polynomial at nuclear scales. TetA is the unbound gravitational field: spacetime itself, the ocean in which matter sits. TetB is the bounded interior: a cavity that can hold confined field content. When stellar compression packs gravity fields into the TetB cavity of a host field, the coupling path traverses the complete Field Origin, crossing $N = 2$ faces. The $N = 2$ constitutive law is

$$x(1 - \mu)^2 = \mu \,, \tag{3}$$

with the physical root

$$\mu_2(x) = \frac{(2x + 1) - \sqrt{4x + 1}}{2x} \,. \tag{4}$$

The nuclear conformal map is $s_2(x) = 2\mu_2(x) - 1$, and the compression ratio is $x = A^2$, where A is the number of gravity fields packed in the host cavity (the mass number).

Substituting $s = s_2(A^2)$ into $f(K, s)$ and subtracting the Field Origin capacity gives the binding energy per field:

$$\boxed{\frac{\text{BE}}{A} = C \cdot \left[\, f\big(K,\ s_2(A^2)\big) - F_{\text{ZERO}} \,\right]} \tag{5}$$

The coupling polynomial evaluated at the nuclear conformal map, minus its value at the Field Origin, is the binding energy. The range of f from $F_{\text{MINUS}} = 13$ to $F_{\text{PLUS}} = 21$ maps to a binding energy range of $C \times (-4)$ to $C \times (+4)$ MeV, with zero binding at $f = F_{\text{ZERO}} = 17$.

2 The energy constant

The energy constant C sets the scale of nuclear binding in MeV. We derive it from the electron mass and the topology.

The electron mass m_e is the unique mass for which the gravity envelope radius coincides with the classical electron radius, derived from K, G, α, $\hbar$, c, and a_0 [1]. The binding energy ceiling is the maximum energy the Field Origin can channel into packing. This is the electron rest energy (the vortex inversion energy at the envelope boundary) multiplied by the gate capacity F_{ZERO}, distributed across K active coupling channels:

$$C = \frac{m_e c^2 \cdot F_{\text{ZERO}}}{K} = \frac{0.511 \times 17}{4} = 2.172 \text{ MeV} \,. \tag{6}$$

The empirical value obtained by fitting to the 82-isotope mean is $C_{\text{fit}} = 2.16$ MeV, a

discrepancy of 0.54%. The asymptotic binding ceiling is

$$\begin{aligned} C(F_{\text{PLUS}}-F_{\text{ZERO}}) &= C \times 4 \\ &= m_e c^2 \cdot F_{\text{ZERO}} \\ &= 8.687 \text{ MeV}\,, \end{aligned} \tag{7}$$

consistent with the observed peak at Fe-56 (BE/A = 8.790 MeV), whose individual element-specific binding exceeds the geometric mean by approximately 1.2% [2].

3 Binding energy validation

Equation (5) with the derived C from Eq. (6) is applied to 82 stable isotopes (He-4 through Bi-209). No parameters are adjusted.

The results are tabulated in Table 1. Summary statistics:

Quantity	Value
Elements tested	82
R^2 (total BE)	0.991
Mean |error| (BE/A)	3.9%
Within 5%	65 (79%)
Within 10%	80 (98%)
Beyond 15%	1 (He-4)
Fitted parameters	0

The single outlier is He-4, whose binding energy per field (7.074 MeV observed vs 4.853 MeV predicted) is systematically underpredicted. He-4 is the first complete $K = 4$ closure ($A = K$) and receives a geometric resonance bonus that the constant-C formula does not capture. Excluding He-4, the mean error across 81 isotopes is 3.5%.

The heavy-element end (Pb, Bi, Tl, Hg) shows a systematic overprediction of 6–10%, attributable to Coulomb repulsion among vortex-active (proton) fields, which grows as Z^2 and is not captured by the A-only formula. The Bethe-Weizsäcker mass formula achieves comparable coverage with five fitted parameters. This formula uses zero.

4 The packing algebra

The binding energy formula determines *how tightly* fields are held. The packing algebra determines *how they arrange*: which configurations are stable, where the magic numbers fall, and why the shell structure has the form it does.

Gravity fields packed inside the tetrahedral cavity (TetB) arrange in three-dimensional tetrahedral close-packing, constrained by the $K = 4$ channel structure of the host. The packing is recursive: once $K = 4$ fields fill the host channels, additional fields pack inside the existing residents, each of which is itself a complete stellated octahedron. At each nesting level, the Field Origin of each host is reserved for structural coupling to its parent (the gate). This gate is not available for packing. The gate exclusion follows the Sierpiński construction: at each triforce level, the central sub-triangle is removed, leaving three outer sub-triangles for further recursion.

Three functions encode the complete shell structure. The tetrahedral capacity is

$$T(n) = \frac{n(n+1)(n+2)}{6}\,, \tag{8}$$

the gate exclusion is

$$G(n) = \begin{cases} 0 & n < K \\ (n-1)\,n/2 & n \geq K \end{cases}\,, \tag{9}$$

and the available positions are

$$P(n) = T(n) - G(n)\,. \tag{10}$$

The magic number at shell n is

$$\boxed{M(n) = 2\,P(n)} \tag{11}$$

where the factor of 2 accounts for the two time-face orientations (spin degeneracy). The packing is subject to the nesting ceiling

$$M(n) \leq K^K = 256\,. \tag{12}$$

Evaluating Eqs. (8)–(11) at $K = 4$:

n	T	G	P	M	S
1	1	0	1	2	2
2	4	0	4	8	6
3	10	0	10	20	12
4	20	6	14	28	8
5	35	10	25	50	22
6	56	15	41	82	32
7	84	21	63	126	44
8	120	28	92	184	58
9	165	36	129	**258**	74

Gate exclusion ($G > 0$) begins at $n = K = 4$. Shell 9 requires $M(9) = 258 > K^K = 256$ and cannot close.

The column $S(n) = M(n) - M(n-1)$ gives the shell size (nucleons per shell). Shell 3 has $S = 12$ and shell 4 has $S = 8$. This inversion ($S_3 > S_4$) is the spin-orbit effect: in standard nuclear physics it requires introducing the spin-orbit force with a fitted coupling constant. Here it is a theorem. Below K, shell sizes follow $n(n+1)$, giving $3 \times 4 = 12$ at $n = 3$. The transition shell at $n = K$ has size $2K = 8$. Since $D \times K = 12 > 2K = 8$ (because $D = 3 > 2$), the inversion is geometrically forced.

Uniqueness. Testing all integers $2 \leq K \leq 10$, only $K = 4$ produces the observed magic number sequence $\{2, 8, 20, 28, 50, 82, 126\}$. $K = 3$ misses 28. $K = 5$ misses 20. The data selects the stellated octahedron uniquely.

5 The stability ceiling

The recursive nesting has maximum depth $K = 4$, corresponding to the $3 + 1$ degrees of freedom of spacetime: three spatial dimensions and one temporal. Each nesting level consumes one degree of freedom. The total addressable state space is $K^K = 4^4 = 256$.

Shell 8 closes at $M(8) = 184 < 256$, safely within the ceiling. This corresponds to the predicted island of stability near $Z \approx 114$, $N = 126$–184. Shell 9 would require $M(9) = 258 > 256$, exceeding the ceiling by exactly 2 (the number of time faces). Shell 9 cannot close. Beyond this point, nuclear configurations are progressively unstable.

Bismuth-209, the heaviest observationally stable isotope, uses $209/256 = 81.6\%$ of the address space. Fermium-258 ($A = 258 = M(9)$) decays by spontaneous fission with a half-life of 370 μs, consistent with complete exhaustion of the nesting hierarchy.

6 Galactic and nuclear scales from one polynomial

The coupling polynomial $f(K, s)$ governs both scales of the framework through a single structural parameter: N, the number of tetrahedral faces on the coupling path.

At galactic scales ($N = 1$, one face crossing), the conformal map is $s_1(x) = (x-1)/(x+1)$ and the constitutive law $\mu_1(x) = x/(x+1)$ is the Möbius map of the polynomial. Its antiderivative

$$F(y) = y - 2\sqrt{y} + 2\ln(1 + \sqrt{y}) \tag{13}$$

is the AQUAL energy functional [8], from which galaxy rotation curves follow without dark matter. At nuclear scales ($N = 2$, full Field Origin traversal), the conformal map shifts to $s_2(A^2) = 2\mu_2(A^2) - 1$, and the polynomial evaluates to

$$\begin{aligned} f\big(K, s_2(A^2)\big) &= F_{\text{MINUS}} + 2K\,\mu_2(A^2) \\ &= 13 + 8\,\mu_2(A^2)\,. \end{aligned} \tag{14}$$

The binding energy is C times the excess above $F_{\text{ZERO}} = 17$.

The two scales share the polynomial, the three landmark values $(13, 17, 21)$, and the integer $K = 4$. They differ only in N. One face crossing for galaxies. Two for nuclei. The same stellated octahedron at both scales.

7 Discussion

What is established. The binding energy formula Eq. (5) with derived C (Eq. 6) reproduces 82 isotope binding energies at $R^2 = 0.99$ with zero fitted parameters. The packing algebra Eqs. (8)–(11) produces all seven magic numbers exactly, with $K = 4$ as the unique solution. The stability ceiling $K^K = 256$ correctly predicts the endpoint of nuclear stability. The spin-orbit inversion is a geometric theorem.

What is not established. The Z-dependent Coulomb correction (vortex repulsion among proton fields) is not separated in the current formula. Incorporating it would require a second term proportional to $Z(Z-1)/A^{1/3}$, reducing the heavy-element overprediction. However, the Coulomb interaction is not additive in this framework: previous analysis shows that subtracting the standard Coulomb term *degrades* the fit, suggesting the electromagnetic contribution is already partially encoded in the constitutive law [2].

The element-specific cavity resonance structure (golden-ratio harmonics within TetB) produces $\sim$5% element-by-element variation in the effective C-factor. This is documented but not included in the present letter.

Reproducibility. A self-contained Python validation script and an interactive periodic table tool are available as supplementary material at [9]. The script requires no external libraries and runs the complete 82-isotope validation in under one second.

Acknowledgments

The author thanks the SPARC collaboration for public rotation curve data and the AME2020 collaboration for nuclear mass evaluations.

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Table 1: GFD matter profile for all 83 stable elements. A is the most abundant stable isotope mass number. Shell n from Eq. (11). μ_2 and s_2 from Eq. (4). BE/A predicted from Eq. (5) with $C = 2.172$ MeV. Observed values from AME2020 [10]. Δ is the ceiling fraction $A/256$.

Z	Sym	A	n	$P(n)$	$M(n)$	μ_2	s_2	BE/A (MeV) GFD	BE/A (MeV) Obs	Err (%)	Δ (%)
1	H	1	1	1	2	0.382	−0.236	0.00	0.00		0.4
2	He	4	2	4	8	0.779	+0.559	4.85	7.07	−31	1.6
3	Li	7	2	4	8	0.867	+0.734	6.38	5.61	+14	2.7
4	Be	9	3	10	20	0.895	+0.790	6.86	6.46	+6	3.5
5	B	11	3	10	20	0.913	+0.826	7.18	6.93	+4	4.3
6	C	12	3	10	20	0.920	+0.840	7.30	7.68	−5	4.7
7	N	14	3	10	20	0.932	+0.865	7.49	7.48	+0	5.5
8	O	16	3	10	20	0.939	+0.879	7.63	7.98	−4	6.3
9	F	19	3	10	20	0.949	+0.898	7.80	7.78	+0	7.4
10	Ne	20	3	10	20	0.951	+0.903	7.84	8.03	−2	7.8
11	Na	23	4	14	28	0.957	+0.914	7.94	8.11	−2	9.0
12	Mg	24	4	14	28	0.959	+0.918	7.97	8.26	−4	9.4
13	Al	27	4	14	28	0.964	+0.927	8.05	8.33	−3	10.5
14	Si	28	4	14	28	0.965	+0.930	8.08	8.45	−4	10.9
15	P	31	5	25	50	0.968	+0.936	8.13	8.48	−4	12.1
16	S	32	5	25	50	0.969	+0.938	8.15	8.49	−4	12.5
17	Cl	35	5	25	50	0.972	+0.943	8.19	8.52	−4	13.7
18	Ar	40	5	25	50	0.975	+0.951	8.26	8.60	−4	15.6
19	K	39	5	25	50	0.975	+0.949	8.24	8.56	−4	15.2
20	Ca	40	5	25	50	0.975	+0.951	8.26	8.55	−3	15.6
21	Sc	45	5	25	50	0.978	+0.956	8.30	8.62	−4	17.6
22	Ti	48	5	25	50	0.979	+0.959	8.33	8.72	−5	18.8
23	V	51	6	41	82	0.981	+0.961	8.35	8.74	−5	19.9
24	Cr	52	6	41	82	0.981	+0.962	8.35	8.78	−5	20.3
25	Mn	55	6	41	82	0.982	+0.964	8.37	8.77	−5	21.5
26	Fe	56	6	41	82	0.982	+0.965	8.38	8.79	−5	21.9
27	Co	59	6	41	82	0.983	+0.966	8.39	8.77	−4	23.0
28	Ni	58	6	41	82	0.983	+0.966	8.39	8.73	−4	22.7
29	Cu	63	6	41	82	0.984	+0.969	8.41	8.75	−4	24.6
30	Zn	64	6	41	82	0.984	+0.969	8.41	8.74	−4	25.0
31	Ga	69	6	41	82	0.986	+0.971	8.43	8.72	−3	27.0
32	Ge	74	6	41	82	0.987	+0.973	8.45	8.71	−3	28.9
33	As	75	6	41	82	0.987	+0.974	8.45	8.70	−3	29.3
34	Se	80	6	41	82	0.988	+0.975	8.47	8.71	−3	31.3
35	Br	79	6	41	82	0.987	+0.975	8.47	8.70	−3	30.9
36	Kr	84	7	63	126	0.988	+0.976	8.48	8.72	−3	32.8
37	Rb	85	7	63	126	0.988	+0.977	8.48	8.70	−3	33.2
38	Sr	88	7	63	126	0.989	+0.977	8.49	8.73	−3	34.4
39	Y	89	7	63	126	0.989	+0.978	8.49	8.71	−3	34.8
40	Zr	90	7	63	126	0.989	+0.978	8.49	8.71	−3	35.2
41	Nb	93	7	63	126	0.989	+0.979	8.50	8.66	−2	36.3
42	Mo	98	7	63	126	0.990	+0.980	8.51	8.64	−1	38.3
43	Tc	98	7	63	126	0.990	+0.980	8.51	8.59	−1	38.3
44	Ru	102	7	63	126	0.990	+0.980	8.51	8.61	−1	39.8
45	Rh	103	7	63	126	0.990	+0.981	8.52	8.58	−1	40.2
46	Pd	106	7	63	126	0.991	+0.981	8.52	8.57	−1	41.4
47	Ag	107	7	63	126	0.991	+0.981	8.52	8.55	0	41.8
48	Cd	114	7	63	126	0.991	+0.982	8.53	8.54	0	44.5
49	In	115	7	63	126	0.991	+0.983	8.53	8.52	0	44.9
50	Sn	120	7	63	126	0.992	+0.983	8.54	8.51	0	46.9
51	Sb	121	7	63	126	0.992	+0.984	8.54	8.48	+1	47.3
52	Te	130	8	92	184	0.992	+0.985	8.55	8.47	+1	50.8
53	I	127	8	92	184	0.992	+0.984	8.55	8.45	+1	49.6

Z	Sym	A	n	$P(n)$	$M(n)$	μ_2	s_2	BE/A (MeV) GFD	Obs	Err (%)	Δ (%)
54	Xe	132	8	92	184	0.992	+0.985	8.56	8.44	+1	51.6
55	Cs	133	8	92	184	0.993	+0.985	8.56	8.41	+2	52.0
56	Ba	138	8	92	184	0.993	+0.986	8.56	8.39	+2	53.9
57	La	139	8	92	184	0.993	+0.986	8.56	8.38	+2	54.3
58	Ce	140	8	92	184	0.993	+0.986	8.57	8.38	+2	54.7
59	Pr	141	8	92	184	0.993	+0.986	8.57	8.35	+3	55.1
60	Nd	142	8	92	184	0.993	+0.986	8.57	8.35	+3	55.5
61	Pm	145	8	92	184	0.993	+0.986	8.57	8.32	+3	56.6
62	Sm	152	8	92	184	0.993	+0.987	8.57	8.30	+3	59.4
63	Eu	153	8	92	184	0.993	+0.987	8.57	8.28	+4	59.8
64	Gd	158	8	92	184	0.994	+0.987	8.57	8.27	+4	61.7
65	Tb	159	8	92	184	0.994	+0.987	8.57	8.25	+4	62.1
66	Dy	164	8	92	184	0.994	+0.988	8.58	8.25	+4	64.1
67	Ho	165	8	92	184	0.994	+0.988	8.58	8.23	+4	64.5
68	Er	166	8	92	184	0.994	+0.988	8.58	8.23	+4	64.8
69	Tm	169	8	92	184	0.994	+0.988	8.58	8.21	+5	66.0
70	Yb	174	8	92	184	0.994	+0.989	8.58	8.20	+5	68.0
71	Lu	175	8	92	184	0.994	+0.989	8.59	8.19	+5	68.4
72	Hf	180	8	92	184	0.994	+0.989	8.59	8.18	+5	70.3
73	Ta	181	8	92	184	0.994	+0.989	8.59	8.17	+5	70.7
74	W	184	8	92	184	0.995	+0.989	8.59	8.15	+5	71.9
75	Re	187	9	129	258	0.995	+0.989	8.59	8.13	+6	73.0
76	Os	192	9	129	258	0.995	+0.990	8.59	8.12	+6	75.0
77	Ir	193	9	129	258	0.995	+0.990	8.59	8.11	+6	75.4
78	Pt	195	9	129	258	0.995	+0.990	8.60	8.10	+6	76.2
79	Au	197	9	129	258	0.995	+0.990	8.60	8.08	+6	77.0
80	Hg	202	9	129	258	0.995	+0.990	8.60	8.06	+7	78.9
81	Tl	205	9	129	258	0.995	+0.990	8.60	8.05	+7	80.1
82	Pb	208	9	129	258	0.995	+0.990	8.60	7.87	+9	81.3
83	Bi	209	9	129	258	0.995	+0.990	8.60	7.85	+10	81.6